Technical Annex:

Instruction manual for conducting climate and disaster risk screening in Malawi

To accompany the Malawi PSIP Preparation Handbook (2025)

Applies to Section 3 of the New Project Proposal (NPP) Template

1 INTRODUCTION

1.1 PURPOSE OF THIS TECHNICAL ANNEX

This technical annex provides step-by-step instructions to support the Government of Malawi's Ministries, Departments, and Agencies (MDAs) in completing Section 3: Climate and Disaster Risk Screening (CDRS) of the New Project Proposal (NPP) Template. It is a mandatory requirement for all public investment proposals submitted for PSIP appraisal.

The Annex ensures that climate and disaster risks are:

- Identified early,
- Assessed consistently, and
- Addressed appropriately in project design,

in line with Malawi's national climate policies, the 2021 Updated Nationally Determined Contribution (NDC), the National Climate Change Management Policy (NCCMP), Malawi Vision 2063 (MW2063), and global good practice in climate-smart public investment management.[1]

1.2 INTENDED USERS

The intended users of this manual are government planning officers and appraisers who play a role in the preparation, submission, and appraisal of public investment proposals. The manual is designed to guide users through the CDRS process in a structured, pragmatic way that supports consistent application across all MDAs.

Targeted users of this manual:

Project Planners (MDAs): These are planners, economists, engineers, sector specialists, and project managers who are responsible for developing new project proposals and completing Section 3 of the NPP Template. The manual helps planners understand:

- What information is required for each question;
- How to apply climate and disaster risk concepts in a project context;
- Where to find data or evidence; and
- How to provide clear and credible justifications that meet appraisal standards.

Project Appraisers (PSIP Unit): These are officers responsible for assessing the completeness, clarity, and credibility of submissions during PSIP screening and appraisal. Appraisers will use this manual to:

- Apply scoring criteria ("Pass", "Not Clear", "Fail") consistently;
- Determine whether risk assessments are aligned with national guidance;
- Identify gaps requiring clarification; and
- Ensure that climate and disaster risks have been adequately considered before projects advance further in the PSIP cycle.

[1] See, for example, the World Bank's Reference Guide for Climate-Smart Public Investment and other resources located here: https://www.worldbank.org/en/programs/climate-change-governance/publications

By clarifying the roles and expectations of both groups, this manual supports a transparent and predictable screening process and strengthens the integration of climate resilience across Malawi's public investments.

Other stakeholders who stand to benefit from this document include Ministry of Finance officers involved in public investment oversight and development partners providing technical or financial support to Malawi's climate-resilient development efforts.

1.3 How to Use this Manual

This manual is structured to guide users through the CDRS section of the NPP in a clear, step-by-step manner. Each screening question is accompanied by detailed instructions, examples, and notes to help both planners and appraisers understand what is required, how to interpret the available response options, and how to ensure alignment with Malawi's climate policies and PSIP standards.

Users are encouraged to read the manual alongside the NPP while completing or assessing the CDRS. Each question includes an explanation of the task, the type of evidence needed, and examples of credible responses. In cases where project information is still evolving, the manual also highlights how to document uncertainty and when to propose follow-up assessments. To support quick reference during preparation and appraisal, the core functions of the manual are summarized in Box 1.

Box 1. What this manual provides

- ✓ A clear explanation of what each question in Section 3 of the NPP is asking.
- ✓ Step-by-step instructions on how to complete each aspect of the CDRS.
- ✓ Key tasks and minimum evidence requirements for project preparers.
- ✓ Guidance on interpreting and selecting response options.
- ✓ Common errors and good-practice examples to ensure credible submissions.
- ✓ Review criteria for PSIP appraisers to support consistent scoring.

2 PART I: FOUNDATIONS

2.1 KEY CONCEPTS AND DEFINITIONS

Climate and disaster risk screening relies on a set of foundational concepts that help project teams identify, understand, and respond to risks that may affect project performance, cost, timelines, and sustainability. This section provides short, practical definitions per IPCC[2] tailored to the Malawi context. These terms are used throughout the NPP and this manual.

- **Adaptive capacity**: The ability of the project (in design, operations, management, or institutional arrangements) to withstand, adjust to, recover from, or take advantage of opportunities arising from the impacts of climate or geophysical hazards.

- **Adaptation**: Measures that reduce vulnerability to climate change, strengthen resilience, or improve long-term project reliability in the face of hazards.

- **Climate:** In a narrow sense, climate is usually defined as the average weather, or more rigorously as the statistical description in terms of the mean and variability of relevant quantities over a period of time ranging from months to thousands or millions of years. The classical period for averaging these variables is 30 years, as defined by the World Meteorological Organization. The relevant quantities are most often surface variables such as temperature, precipitation and wind. Climate in a wider sense is the state, including a statistical description, of the *climate system*.

- **Climate change:** A change in the state of the climate that can be identified by changes in the mean and the variability of its properties and that persists for an extended period, typically decades or longer. Climate change may be due to natural internal processes or external forcings such as modulations of the solar cycles, volcanic eruptions and persistent anthropogenic changes in the composition of the atmosphere or in land use.

- **Climate hazard**: A potentially damaging climatic event or trend, such as floods, droughts, heatwaves, storms, or wildfires that may cause loss of life, injury, or other health impacts, as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems and environmental resources. Hazards can occur suddenly or intensify over time due to climate change.

- **Climate proofing:** A process that involves identifying and integrating climate and disaster risks and opportunities in the design, implementation, operation and maintenance of development projects

[2] See IPCC Glossary (https://apps.ipcc.ch/glossary/)

- **Climate projection**: Simulated response of the climate system to a scenario of future emissions or concentrations of greenhouse gases (GHGs) and aerosols and changes in land use, generally derived using climate models.

- **Climate risk:** The potential for adverse consequences for human or ecological systems, recognizing the diversity of values and objectives associated with such systems. Risks can arise from potential impacts of climate change as well as human responses to climate change. Relevant adverse consequences include those on lives, livelihoods, health and well-being, economic, social and cultural assets and investments, infrastructure, services (including ecosystem services), ecosystems and species.

- **Exposure**: The location-based dimension of risk. Exposure describes whether the project site, its assets, its supply chains, or its beneficiaries are situated in areas where hazards occur or are expected to occur.

- **Geophysical hazards**: Naturally occurring events such as earthquakes, landslides, and volcanic activity. These hazards are not as influenced by climate variables or human action, and the potential occurrence of an event can span over many centuries. Although not climate-driven, they are included because they pose material risks to infrastructure and communities.

- **Impact**: Effect on natural and human systems.

- **Indicator**: Quantitative, qualitative, or binary variable that can be measured or described, in response to a defined criterion.

- **Mitigation** (of climate change): A human intervention to reduce greenhouse gas emissions or enhance carbon sinks.

- **Residual risk**: The level of risk that remains after all existing and planned adaptation measures have been implemented. It reflects the fact that no project can eliminate risk entirely.

- **Resilience**: The capacity of interconnected social, economic and ecological systems to cope with a hazardous event, trend or disturbance, responding or reorganizing in ways that maintain their essential function, identity and structure. Resilience is a positive attribute when it maintains capacity for adaptation, learning and/or transformation.

- **Risk assessment:** The qualitative and/or quantitative scientific estimation of *risks*.

- **Risk management** (mitigation of project risks): Plans, actions, strategies or policies to reduce the likelihood and/or magnitude of adverse potential consequences, based on assessed or perceived risks.

- **Sensitivity**: The degree to which a project's systems, components, or services are susceptible to harm when a hazard occurs. Sensitivity reflects the characteristics of the

project itself, not its location.

- **Vulnerability**: The propensity or predisposition to be adversely affected. Vulnerability is a function of exposure, sensitivity, and adaptive capacity. A highly exposed and sensitive project with low adaptive capacity is more vulnerable to climate and disaster risks.

These concepts form the backbone of the CDRS that is presented in the NPP. They help ensure that climate and disaster risks are considered systematically and realistically across all public investment proposals.

2.2 DATA SOURCES AND REFERENCE MATERIALS

Risk screening does not require advanced technical modelling. It does, however, require the use of credible and accessible data sources to justify exposure ratings, sensitivity assessments, and climate-related impacts. Project teams should rely on a combination of national and international resources, including:

National sources	International sources (used when national data is limited or unavailable)
• Department of Climate Change and Meteorological Services • Malawi Meteorological Department • Ministry of Natural Resources and Climate Change • Basin, district, or sector hazard maps • Environmental and Social Impact Assessments • Feasibility studies or engineering assessments	• ThinkHazard (Global hazard screening) • IPCC regional information • UNDRR/Global Risk Data • World Bank Climate Change Knowledge Portal • Academic or technical reports from reputable institutions

Where data is not available, professional judgement, supported by short explanations, may be used. For example, planners with long experience in local flood-prone areas can cite observed trends. However, general statements ("the area sometimes floods") are not sufficient; specific observations and sources should be noted.

3 PART II: STEP-BY-STEP GUIDE FOR COMPLETING THE CDRS

The NPP Template guides MDAs through a structured assessment of the project's alignment with national climate objectives, its exposure to climate and geophysical hazards, its vulnerability, and the adequacy of proposed mitigation and adaptation measures. Part II of the instruction manual provides question-by-question guidance to help users complete the assessment accurately, consistently, and in a manner that strengthens the project's justification and long-term viability. For each question in Section 3 of the NPP, this part explains:

- What the question seeks to understand
- How project teams should approach it
- What evidence or reasoning supports a credible response, and
- What appraisers look for when applying the scoring criteria.

This section follows the exact flow of the NPP Template and is designed to be read alongside it.

3.1 CLIMATE POLICY CONTEXT

This section of the CDRS clarifies how the project aligns with Malawi's national climate strategies, including the NCCMP, Updated NDC, and MW2063 implementation priorities. These questions help establish whether the project is climate-motivated and whether it meaningfully contributes to mitigation or adaptation objectives.

CDRS Question 3.1a: Is climate change a principal or significant motivation for the project?

Purpose of the Question

This question determines whether climate change concerns are among the leading reasons for proposing the project. A "principal" motivation means the project is directly designed to address a climate-related problem or opportunity. An activity may be marked principal when the objective (climate change mitigation or adaptation) is explicitly stated as fundamental in the design of, or the motivation for, the project. Promoting the objective will thus be stated in the project documentation as one of the principal reasons to undertake it. In other words, the project would not have been undertaken (or designed that way) but for that objective.

A "significant" motivation indicates that climate considerations materially influence design choices, even if the primary purpose is not climate-related. A project may be marked significant when the objective (climate change mitigation or adaptation) is explicitly stated but it is not the fundamental driver or motivation. Instead, the project has other prime objectives but has been formulated or adjusted to help meet relevant climate concerns.
A project may be marked no / not applicable when it does not target climate change objectives (mitigation or adaptation) in any significant way.

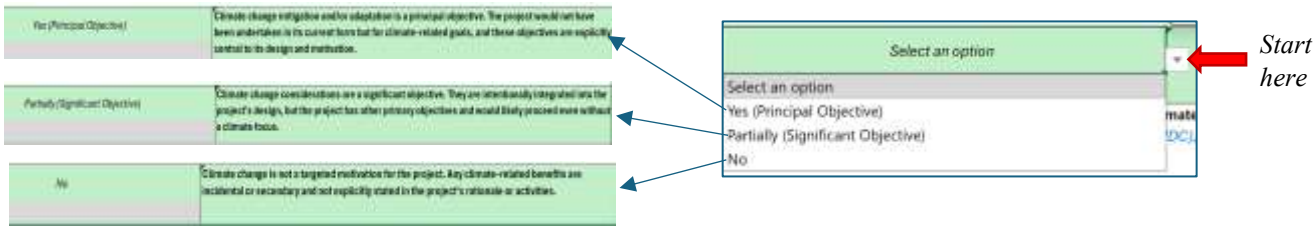
Completing this task will provide project teams with an opportunity to tag projects for climate expenditure tracking purposes, enabling reconciliation with climate budgeting tagging initiatives. Projects that identify climate change as a principal motivation typically contribute to fulfilling policy commitments to lessen GHG emissions (mitigation), build resilience to the anticipated effects of climate change (adaptation), or facilitate transition to a climate changed world. Initiatives range from pure mitigation projects that focus on carbon capture, to pure adaptation projects like infrastructure for flood protection. They may also include investments in climate proofing to protect existing or planned assets from climate change hazards.

SIDE NOTE: It is reasonable to anticipate that all projects, regardless of motivation, will be affected by climate and disaster risks to varying degrees. Thus, standard investment projects may need to build in risk mitigation measures to minimize impacts of anticipated events. For example, a road project exposed to an increasing risk of flooding can increase resilience by incorporating a drainage system to remove excessive water, reducing potential damage and traffic interruptions.

How project planners should respond

- Review the project's core objectives and rationale.
- Identify whether climate threats (e.g., floods, drought, heat, storms) or opportunities (e.g., low-carbon transition, renewable energy access) are key drivers.
- If climate considerations meaningfully influence objectives, design, or benefits, select the appropriate option.

As illustrated below, the dropdown menu will offer three responses: "Yes (Principal Objective)", "Partially (Significant Objective)", and "No". Selecting any option will automatically generate a short explanation. Project planners need only to verify that the explanation for the chosen option aligns with the project's core objectives and rationale.



What appraisers look for

- A response that is directly linked to the problem statement in Section 1 of the NPP.
- Internal consistency between this answer and responses in 1.1, 3.1b, 3.4, and 3.5.

The question may be scored as "Not Clear" if the selected option is not credible or requires clarification based on information provided elsewhere in the NPP or other project documents. The question will be scored "Fail" if there was no attempt to answer the question.

CDRS Question 3.1b: How will the project contribute to Malawi's climate change objectives?

Purpose of the Question

This question determines how the project aligns with Malawi's national climate change commitments by requiring project teams to select, in sequence, the project's climate action, climate objective, relevant sector, and specific NDC-aligned measure. The cascade ensures that contributions are:

- Grounded in official national climate policy, including the NCCMP, the Updated NDC, and sector strategies;
- Consistent with the climate logic of the project, as described in Section 1 (Problem Statement) and Section 3.1a (Motivation);
- Described in a standardized and comparable format suitable for national climate reporting and project tagging.

Completion of this question also supports climate finance tracking, climate budget tagging, and verification of whether public investment proposals advance national adaptation or mitigation priorities.

How project planners should respond

Project planners should follow the four-step cascade exactly as presented in the NPP Template:

1. First, select the type of climate action the project supports (mitigation, adaptation, no significant contribution). Notice that if "no significant contribution" is selected, no response is needed for items (ii), (iii), or (iv).
2. Next, select the national climate objective to which the project is aligned, based on the Updated NDC or NCCMP.
3. Afterwards, choose the climate priority sector most directly relevant to the climate change actions being proposed by the project.
4. Finally, select the specific climate-related measure that the project will directly finance, deliver, or enable.

See Box 2 for a set of decision rules that may be applied for interpreting the response options.

After completing the cascade, the "Explanation" field must include a brief, project-specific justification describing:

- What the project will do,
- Who or what is affected,
- Which hazard or emissions source is addressed, and
- How the action supports the selected NDC or NCCMP measure.

A strong explanation accomplishes the following:

- It cites the exact part of the national policy framework that is relevant (e.g., Updated NDC, Section 6, Water Sector measure).

- It refers to a concrete project activity (e.g., storage tanks, solar pumps, early warning systems).
- It names the population or system benefiting.
- It links design features to the climate objective.

Box 2: Decision rules for interpreting the response options for Q3.1b

1. Select **"Adaptation"** when the project reduces exposure, sensitivity, or improves adaptive capacity to climate hazards.

2. Select **"Mitigation"** when the project reduces GHG emissions, avoids future emissions, or increases carbon sequestration.

3. Select "**Enabling conditions for mitigation**" or "**Enabling conditions for adaptation**" when the project contributes to the NDC's cross-cutting priorities, especially actions that strengthen policies, institutions, governance systems, data and information systems, or finance mechanisms that support climate action, even if the project does not directly build physical assets.

4. **Selecting the correct sector**: Choose the sector representing the dominant climate-related investment or activity. If multiple sectors are touched by the project, select the one most directly financed.

5. **Selecting the specific measure**: Select the NDC-aligned measure that best captures the main climate-related activity financed by the project. If unclear, choose the measure where the project contributes most substantially and directly, not the one where benefits are incidental.

6. Select "**Other (specify)**" only when the project supports actions that do not fall under any of the pre-defined categories.

Mistakes to avoid

- Selecting multiple cascade paths by choosing measures inconsistent with the sector.
- Describing intended future benefits instead of concrete financed activities.
- Using generic statements such as "the project contributes to adaptation."
- Copying extract text from the NDC without specifying how the project contributes.
- Selecting an NDC action because it appears desirable, not because the project supports it.

Example of a strong explanation (for a theoretical selection):

3.1b How will the project contribute to the Republic of Malawi's climate change objectives? *(In responding, ensure consistency with the National Climate Change Management Policy, Malawi's Nationally Determined Contribution (NDC), and other relevant national climate policy documents. Where applicable, explicitly reference how the project aligns with these frameworks.)*		
Selection	**Contribution**	**Explanation**
Adaptation	*<-- First, select the climate change action the project supports*	The project directly supports Malawi's water-related adaptation priorities by increasing access to reliable water resources for drought-affected rural households in Ntcheu, Balaka, and Machinga Districts. By financing the construction of community-level water harvesting systems and small-scale storage structures, the project reduces exposure to recurrent droughts and strengthens the adaptive capacity of vulnerable populations whose livelihoods depend on rain-fed agriculture. This aligns with the Water Sector adaptation measures in Malawi's Updated NDC, which prioritise enhanced water harvesting and storage to build resilience in drought-prone areas.
Plan and implement adaptation actions toward an increased resilience of the most vulnerable Malawians	*<-- Then, select the climate change objective this project targets*	
Water	*<-- Next, select the climate priority sector most relevant to this project*	
***Water harvesting and storage in drought-prone areas	*<-- Finally, select the specific climate-related measure to be supported*	

What appraisers look for

- A credible causal link between the selected measures and the actual project activities.
- Selections that match the project's scope and design, not aspirational language or unsubstantiated claims.
- Internal consistency, especially with Sections 1.1, 3.1a, 3.4, and 3.5.
- A concise explanation referencing policy documents where possible national climate documents or sector strategies.

Scoring triggers:

- **Pass:**
  - Cascade selections reflect actual project activities; AND
  - The explanation is specific, credible, and policy-consistent.

- **Not Clear:**
  - Selection is plausible but lacks adequate justification OR
  - Explanation is generic or not traceable to project activities.

- **Fail:**
  - Selections contradict the project design; OR
  - No explanation is provided; OR
  - Measures selected are impossible for the project to deliver.

3.2 VULNERABILITY ASSESSMENT (EXPOSURE, SENSITIVITY, ADAPTIVE CAPACITY)

This subsection evaluates how climate and geophysical hazards may affect project performance, cost, or sustainability over its lifespan. It is a core part of screening and informs the design of adaptation measures in Section 3.3.

The assessment is structured around three dimensions:

1. **Exposure**: What hazards the project faces based on location.
2. **Sensitivity**: How susceptible the project's components are to those hazards.
3. **Adaptive Capacity**: How well the project can cope with hazard impacts.

CDRS Question 3.2a [Exposure]: What hazards affect the project site?

Purpose of the Question

This question identifies which climate and geophysical hazards may affect the project because of where it is located, both current and in the future. The output is a set of hazard-specific exposure ratings (No/Low, Moderate, High) that are grounded in spatial evidence such as local hazard profiles, historical events, and authoritative climate information. The exposure assessment serves four functions:

1. Locate the project within Malawi's hazard landscape (flood zones, drought-prone districts, heat-stress corridors, steep slopes, cyclone tracks, etc.).
2. Determine which hazards must be linked to sensitivity ratings in Question 3.2b and which require risk-reduction measures in Question 3.3.
3. Ensure internal consistency with Section 1 (Problem Statement), 3.1a (Motivation), 3.1b (Cascade), 3.4 (Risk Implications), and 3.5–3.6 (Strategies).
4. Provide a factual foundation for adaptation tagging and for any claims made regarding climate risk reduction.

Exposure is location-dependent, not design-dependent. A facility in a floodplain has high flood exposure regardless of whether it is well-designed or poorly designed. Exposure does not assess the project's vulnerability (that is covered under Sensitivity and Adaptive Capacity).

How project planners should respond

- Consult available hazard data (national sources preferred; reputable international tools such as ThinkHazard are acceptable).
- Identify all hazards that may affect the site during the project's operational life.
- Assign each hazard an exposure rating (No/Low, Moderate, or High).
- Provide a brief justification linking the rating to a data source, past event, or observed trend.

Project planners should conduct a simple, evidence-linked hazard scan using the following steps:

1. **Check authoritative national data first**
   - DoDMA District Risk Profiles
   - Department of Climate Change and Meteorological Services (DCCMS) climate reports
   - Ministry of Water, Roads Authority, Forestry Department sector hazard maps

2. **Complement with reputable international tools as needed**
   - ThinkHazard
   - IPCC regional information
   - World Bank Climate Change Knowledge Portal

   (These are accepted but should not override national sources.)

3. **Identify all hazards relevant to the geographic location of the project**. Typical hazards include flooding, drought, heat stress/extreme temperature, landslides, and wildfires.

4. **Assign an exposure rating for each hazard**. Use the NPP's three-point scale (see Table 1) and link the rating to a specific geographic fact, such as:
   - District lies in recurrent flood-affected zone
   - Site is within meters of river/stream
   - Area records >15–20 drought years per 30-year period
   - Location frequently reaches >34–35°C in hot season
   - Slope >20–30%
   - Historical cyclone wind damage in southern districts

Table 1. Exposure rating descriptions

Rating	Description
No / Low	The component is **rarely / infrequently exposed** to the hazard, with no recorded instances or only isolated occurrences over long historical periods. Future projections suggest little to no increase in hazard frequency.
Moderate	The component is **occasionally exposed** to the hazard, with a moderate frequency of past occurrences. Future projections may indicate increasing hazard frequency.
High	The component is **frequently exposed** to the hazard, with regular occurrences. Future projections indicate a continued or worsening trend in hazard frequency.

5. **Provide a short justification**. A justification is required only for hazards rated Moderate or High. Each justification must:
   - Cite the data source, historical events, observed local trends, and future projections (e.g., ThinkHazard ratings, DCCMS reports, DoDMA flood records, Geological Survey bulletins).
   - Use concrete, location-specific evidence such as known flood events, proximity to rivers, elevation, fire-prone hillsides, or documented heat extremes (e.g., "Blantyre experienced severe flooding in 2015 and 2023").
   - Avoid generic statements ("Malawi is exposed to floods"). All text must refer to the actual project site.

- Keep the justification to 1–3 lines per hazard, focusing only on facts relevant to project exposure.

The below screenshot provides an example of hazard exposure assessment for a hypothetical water and sanitation project in Blantyre City.

Hazard	Exposure Rating	Capture your analysis here
1. Flood (river or urban)	High	According to ThinkHazard, Blantyre has HIGH flood risk. It has a history of severe urban flooding, including the January 2015 and March 2023 events that inundated the Limbe–Mudi catchment (DoDMA 2015; DoDMA 2023). With the project site <300 m from the Mudi River and short-duration rainfall projected to intensify (IPCC AR6, 2021), the works face high risk of inundation and drainage overflow.
2. Extreme Heat	Moderate	ThinkHazard classifies Blantyre as MODERATE heat hazard. Blantyre now records 20–30 days/year above 34–36°C, with extremes during the October 2023 southern Africa heatwave (DCCMS 2024), and temperatures are projected to rise further under 2030–2050 scenarios (IPCC AR6). These conditions pose thermal stress to pumping and treatment units.
3. Wildfire	High	ThinkHazard ranks Blantyre's peri-urban hills (e.g., Soche Hill, Ndirande) as HIGH wildfire risk. Dry-season fires are common around Soche, Ndirande, and Mpingwe Hills, with notable outbreaks in 2019 and 2022 (Blantyre City Council Environmental Reports; media archives). Hotter, drier seasons projected for southern Malawi (IPCC AR6) increase risks to surface assets and access routes.
4. Volcano	No / Low	
5. Earthquake	Moderate	ThinkHazard lists Blantyre as MODERATE earthquake risk due to proximity to the Shire Rift Zone which experiences periodic tremors, including the 2018 Mulanje–Chiradzulu event (Geological Survey Department, 2018). Future seismic activity may impose structural stress on reservoirs, tanks, and elevated pipelines.
6. Landslide/Mudslide	No / Low	
7. Cyclone / Severe Storm	No / Low	
8. Drought/Water Scarcity	No / Low	
9. Other (Specify)	No / Low	

What appraisers look for

Appraisers assess whether the exposure rating is:

- Location-based and evidence-linked (local hazard profile, historical events, physical geography).
- Consistent with the project's siting, sector, and description.
- Internally aligned with Sensitivity (3.2b), Risk Implications (3.4), and Strategies (3.5–3.6).
- Accurate in distinguishing exposure from sensitivity.
- Reasonable over the project lifetime (not only current but also plausible future hazard conditions).

A submission will be marked "Not Clear" if:

- Ratings do not match known district conditions.
- No justification is provided.
- Only generic language is used.
- Multiple hazards marked "Unknown" despite readily available data.

A "Fail" is issued if:

- The question is not answered;
- Ratings contradict reliable hazard information;
- Exposure is intentionally misrepresented to avoid identifying climate risks.

CDRS Question 3.2b [Sensitivity]: How sensitive is the project to these hazards?

Purpose of the Question

This question assesses how susceptible each project component is to damage or disruption if the hazards identified in Question 3.2a actually occur. Unlike Q3.2a, which evaluates exposure based on location, Q3.2b is about the project's own characteristics: its materials, design, equipment, operating needs, and external dependencies.

Notice that sensitivity reflects project-specific factors only, not site conditions. A high exposure location does not automatically imply high sensitivity; nor does a resilient design guarantee low sensitivity. The objective is to identify which parts of the project are most vulnerable because of their characteristics, so that adaptation measures in Section 3.3 can be targeted effectively.

A "weakest-link" rule applies for this question. If even one hazard could significantly impair a component now or in the future, that component's sensitivity should be rated High, regardless of how it fares against other hazards. Overall, a strong response should demonstrate:

- Hazard-by-component analysis
- Clear reasoning about specific project features
- Consideration of future hazard severity, not just current conditions
- Logical connections between hazard effects and component vulnerability
- Sufficient detail to support scoring and later adaptation planning

Approach for project planners

The project planner's response should follow the following five steps.

Step 1: Break the project into the four required components. The planner should use the exact categories from the NPP template:

1. Physical infrastructure and onsite assets
2. Inputs needed for operation
3. Networks and linkages to external systems
4. Outputs produced by the project

For each component, briefly identify the key elements included within it (e.g., pumps, control cabinets, treatment units, storage tanks, pipes, chemicals, grid power, access road). These should be described directly within the justification text in the "Cause of Vulnerability" cell. Naming these elements inside the justification helps to clarify what could be affected and why. This information is needed to support credible, component-level sensitivity ratings.

Step 2: Carry forward only relevant hazards from Q3.2. Assess only hazards rated Moderate or High in Q3.2a. Do not include hazards with Low/No exposure.

Step 3: Explain the mechanism of impact for each selected hazard. For each component, state how each hazard would impair materials, equipment, operations, dependencies, or the reliability of outputs. Keep the focus on project-specific features, not the project's location.

Step 4: Select a sensitivity rating using the table below. Choose No/Low, Medium, or High for each component, based on the descriptions and materiality analysis (partial vs. complete interruptions; short vs. prolonged downtime).

Table 2. Sensitivity rating descriptions

Rating	Description
No / Low	The component is **minimally sensitive**, showing inherent resistance or little to no functional dependence on the physical effects of the hazards. Performance remains largely unaffected with little to no material disruptions, even in extreme events. There is no logical mechanism linking the hazard to component impairment.
Moderate	The component is **moderately sensitive**. Hazards may disrupt operations or influence performance, but impacts are limited, manageable, or non-structural. Some resilience features exist, but not enough to eliminate risk. It tolerates some variability, but prolonged or extreme events may cause operational challenges. Impacts are likely to cause delays, efficiency losses, or additional operating costs rather than major damage.
High	The component is **highly sensitive**, with strong functional dependence on climatic conditions. The component depends on fragile, specialized, or climate-sensitive technology. A single hazard could severely disrupt or damage the component. Even moderate variations can cause significant disruption or a full stop. The component lacks redundancy. Future conditions could plausibly elevate impacts to a high level. Proactive adaptation measures are likely required.

Note: When choosing a rating, state the materiality: whether the impact is partial or total, short or prolonged, and why.

Step 5: Apply the "weakest-link" rule and record a brief, component-specific justification. If any one hazard could severely disrupt functionality, rate that component High and state this explicitly. For example, if floods cause only minor damage (Low), but extreme heat could cause critical equipment failure (High), the component sensitivity must be rated High. Base the rating on the severity of potential impacts, not the probability of occurrence (probability is covered in exposure).

Each justification must include which hazards matter, **and** how those hazards would materially affect the component. Material impacts may include physical damage**,** interruption of essential inputs**,** reduced productivity**,** failure rates**,** safety concerns, and operational downtime, among others. Planners should keep explanations clear, factual, specific to the project design, and focused on mechanisms of impact. Avoid generic statements (e.g., "floods can cause damage").

Minimum evidence to record

- Component-specific characteristics that influence sensitivity. The key elements within each component must be named inside the justification.

- Mechanisms of impact for each selected hazard. Include an explanation of how each identified hazard physically affects the component, with a clear logic linking the hazard's effect to potential functional impairment.
- Clear indication of materiality (partial vs. full interruption; likely duration)
- Any relevant design specifications or operating constraints that influence sensitivity

Project planners are encouraged to use a variety of evidence sources, which may include but is not limited to design documents, technical specifications, feasibility or engineering reports, local operational experience, or expert input.

BOX 3: How to interpret and select each rating for Q3.2b (examples)

Physical infrastructure and onsite assets

- **High:** "Electrical control panels and water pumps are highly sensitive to inundation; a 0.5–1.0m flood depth would cause complete failure and require replacement."
- **Moderate:** "Administrative buildings can withstand moderate rainfall but lack adequate roof drainage; heavy storms may cause localized leaks and downtime."
- **Low:** "Storage tanks are elevated and constructed with corrosion-resistant materials, limiting sensitivity to flood and heat hazards."

Inputs needed for operation

- **High:** "The project relies on grid electricity; severe storms or lightning events frequently disrupt supply, halting operations entirely."
- **Moderate:** "Raw materials can be stockpiled, but prolonged transport disruptions during heavy rains could delay operations."
- **Low:** "Key inputs (e.g., software licenses) are digital and unaffected by climate hazards."

Networks and linkages

- **High:** "Project outputs depend on a single access road in a landslide-prone area; slope failure would immediately prevent deliveries."
- **Moderate:** "Telecommunications outages during extreme weather may limit communications but do not stop basic operations."
- **Low:** "The project maintains multiple redundancies for data connectivity, keeping sensitivity low."

Outputs produced by the project

- **High:** "Agricultural extension services rely on field demonstrations that are highly sensitive to extreme heat and flooding, disrupting delivery to farmers."
- **Moderate:** "Public information sessions may be rescheduled during storms; impact is moderate."
- **Low:** "Digital outputs delivered online are minimally affected by climate hazards."

Mistakes to avoid

- Repeating location details (belongs in Q3.2a, not here)
- Assigning ratings without explaining how hazards would affect the component
- Ignoring future conditions (for example, hotter heatwaves, more intense rainfall)
- Not applying the weakest-link rule
- Describing existing or planned adaptation measures here (those belong in 3.3a and 3.3b)

Here is a screenshot of the completed sensitivity assessment for the hypothetical water and sanitation project introduced in Q3.2a.

Project Component	Sensitivity Rating	Cause of vulnerability (capture your analysis here)
Physical infrastructure and onsite assets: *All tangible assets required for project's operation, including buildings, facilities, equipment, machinery, stored materials, and operational processes located at or directly controlled by the project site.*	High	Flooding (high sensitivity). The project includes pumps, electrical panels, treatment units, small storage tanks, and above-ground pipes. These items are highly sensitive to floodwater, because electrical systems and pump motors fail when submerged, and flood-carried mud and debris damage moving parts. Wildfires (high sensitivity). Materials used for pipes, cables, seals, and equipment casings are easily damaged by high heat from nearby fires. Extreme Heat (moderate sensitivity). High temperatures may place extra strain on electrical equipment and cause treatment chemicals to degrade faster. Earthquake (moderate sensitivity). Ground movement from an earthquake can crack concrete, shift equipment, or damage joints and fittings. Since flooding or wildfire could completely shut down the project's physical infrastructure, overall sensitivity is **High**.
Inputs needed for operation: *These are essential resources the project depends on to function, such as energy, water, raw materials, data, or specialized services, regardless of whether they are sourced onsite or externally.*	High	Flooding (high sensitivity). The project depends on electricity, treatment chemicals, raw water, and replacement parts. Flooding can interrupt electricity supply and prevent delivery of chemicals and equipment, which immediately stops pumping and treatment processes. Wildfires (high sensitivity). Fires can cause power outages and restrict access for staff and supply deliveries, making it difficult to maintain treatment and pumping operations. Extreme Heat (moderate sensitivity). High temperatures cause treatment chemicals to degrade faster and place additional strain on motors and electrical equipment. Earthquake (moderate sensitivity). Ground shaking may disrupt raw-water intakes, affect the stability of stored chemicals and equipment, or interrupt external supply chains. Flooding or wildfire can severely interrupt essential operational inputs needed for treatment and water delivery. Therefore, the overall sensitivity is deemed **High**.
Networks and linkages to external systems: *These are connections the project relies on to interact with broader systems, including transportation routes, utilities (power, water, telecommunications), supply chains, distribution networks, and digital or institutional interfaces.*	High	Flooding (high sensitivity). The project relies on a single access route for staff, chemicals, and equipment. Flooding can make this route unusable, preventing essential maintenance and supply deliveries. Floodwater may also affect power or communication services the project depends on. Wildfires (high sensitivity). Fires can lead to power cuts, damage above-ground pipes or connections, and temporarily restrict access for safety reasons. Extreme Heat (moderate sensitivity). Heat increases the likelihood of interruptions to communication services used for monitoring and controlling treatment processes. Earthquake (moderate sensitivity). Earthquakes can damage pipelines, joints, and other external systems that the project relies on for distribution and communication.
Outputs produced by the project: *These are goods, services, or other deliverables generated by the project and supplied to end users, beneficiaries, or downstream systems.*	High	Flooding (high sensitivity). Flood-related equipment failures or treatment overloads can halt the production and distribution of safe water. Service quality may also decline if sediment or debris overwhelms treatment steps. Wildfires (high sensitivity). Power interruptions and access restrictions during fires reduce water production and delay necessary operations, affecting the availability and reliability of treated water. Extreme Heat (moderate sensitivity). High temperatures cause disinfectant levels in the distribution network to drop faster, requiring more frequent adjustments to maintain water safety. Earthquake (moderate sensitivity). Earthquakes can cause leaks or pressure drops in the pipe network, resulting in service interruptions or boil-water notices. Since multiple hazards, especially flooding and wildfire, can severely disrupt the delivery of treated water, overall sensitivity is **High**.

(Hazards carried forward from Q3.2a: Flooding = High; Extreme Heat = Moderate; Wildfire = High; Earthquake = Moderate)

What appraisers look for

- Clear, component-by-component logic focused on project characteristics
- Explanations that specify *how* hazards affect components.
- Evidence that only Moderate/High exposure hazards from Q3.2a were assessed
- Credible judgments of materiality (what stops the system, what only slows it)
- Internal consistency with Q3.2a, 3.3a/3.3b, and 3.4–3.5
- Avoidance of generic statements (e.g., "assets are sensitive to flood").

Scoring guidance

- **Pass:** Each component has a rating with a clear, project-specific justification that reflects materiality and the weakest-link rule.

- **Not Clear:** Ratings are plausible but justifications are generic or incomplete, or future conditions were not considered.

- **Fail:** Ratings have no explanation, rely on location (exposure) rather than project characteristics, ignore the weakest-link rule, or contradict other sections.

CDRS Question 3.2c [Adaptive Capacity]: As currently designed, how well can each project component withstand, adjust to, absorb, or recover from material impacts from the hazards identified in Q3.2a, based on the upper bound of hazard severities that may reasonably occur?

Purpose of the Question

This question assesses how well the project, as it is currently designed, can cope with the full range of climate and geophysical hazard impacts identified in Q3.2a. Unlike sensitivity (Q3.2b), which evaluates how easily components are harmed, adaptive capacity evaluates the ability to withstand harm or bounce back. Adaptive capacity ratings are based on the severity of potential impacts, not the probability of occurrence, which is assessed under exposure (Q3.2a). It captures:

- inherent resilience features (e.g., robust materials, design redundancy, backup systems),
- absorptive capacity (e.g., drainage, structural reinforcement),
- operational and maintenance systems that support continuity,
- institutional, managerial, and financial capacity to respond to disruptions.

Adaptive capacity must be assessed per component, using the four standard project components required by the NPP. Adaptive capacity does not include planned future improvements. Those belong in Q3.3b.

How project planners should respond

Completing the adaptive capacity assessment requires following the five steps outlined below.

Step 1: Carry forward hazards from Q3.2a. Only consider hazards rated Moderate or High exposure in Q3.2a. Ignore hazards with Low/No exposure.

Step 2: Consider the upper-bound severity of each hazard. Adaptive capacity ratings must reflect the worst credible intensity of each hazard (e.g., 1–2 m flood depth, 5–10 days of heat >35°C, severe wildfire conditions), not only typical events.

Step 3: Identify existing design, operational, or institutional features that influence adaptive capacity. Examples of such features include:

- *Structural robustness*: reinforced concrete, elevated equipment, wind-resistant roofing, deep foundations.
- *Redundancy*: backup pumps, additional power sources, multiple supply routes.
- *Flexibility*: modular equipment that can be isolated or repaired without full shutdown.
- *Protective systems*: drainage, surge protection, fire-retardant materials.
- *Institutional readiness*: maintenance protocols, trained staff, rapid-response capacity, emergency standard operating procedures.
- *Financial or resource buffers*: maintenance budgets, inventory stockpiles.

Document only features that already exist in the current design or operation.

Step 4: Select an adaptive capacity rating. Use the NPP's three-point scale that is elaborated in the following table. Unlike exposure and sensitivity, **adaptive capacity is a positive attribute: stronger capacity indicates greater resilience**.

Table 3. Adaptive capacity rating descriptions

Rating	Description
Strong	The component can tolerate upper-bound hazard impacts with limited disruption. Strong resilience features exist, including redundancy, robust materials, and established operations and maintenance protocols. Recovery is likely to be rapid and would not require major external support.
Moderate	The component can tolerate some impacts but not all. Damage or disruption may occur under severe conditions, but the system can recover if repairs or temporary workarounds are applied. Some resilience features exist, but gaps remain.
Weak	The component has little capacity to withstand or recover from upper bound hazard impacts. No meaningful resilience features exist. Severe disruption or extensive repair is likely.

Step 5: Provide a clear, component-specific justification. Each justification must:

- Name the specific component elements (e.g., pump houses, data servers, access roads),
- Identify which hazards matter and why,
- Explain how current design features strengthen or weaken resilience, and
- State whether recovery is partial/complete and short- or long-duration.

Do not mention planned improvements as these belong in Q3.3b.

Minimum evidence required

- Specific design or institutional features supporting the adaptive capacity rating,
- A statement describing how the component responds to each relevant hazard, and

- Evidence-based reasoning linking features to resilience (e.g., "elevated above flood height", "backup generator provides 48 hours autonomy", "drainage designed for 1-in-20-year storm").

Common mistakes to avoid

- Repeating **site exposure** information (belongs in 3.2a)
- Listing measures without judging whether they are sufficient at the upper bound
- Ignoring recovery time and service continuity in the explanation
- Assigning ratings without naming concrete features, redundancies, or buffers
- Counting planned measures here (planned measures belong in 3.3b)

Avoid generic statements such as "the system is resilient" or "capacity is high."

Box 4 provides structured examples that planners may use as templates for drafting their own responses.

BOX 4: Component-by-component guidance and sample justifications for Q3.2c

(This example assumes typical baseline features; replace with project-specific facts in your NPP.)

Physical infrastructure and onsite assets

- **Moderate:** "The main treatment building is constructed with reinforced concrete and elevated 0.8 m above surrounding grade, reducing vulnerability to moderate floods but not to upper bound events (>1 m). Electrical control panels are mounted 1.5 m above floor level and enclosed in moisture sealed cabinets, enabling rapid recovery after moderate inundation. However, no fire hardening features exist, and roof structures are not rated for severe wind loads. Overall, the component can recover from moderate disruptions but would sustain significant damage under severe flood or wind scenarios."

Inputs needed for operation

- **Moderate:** "The project relies primarily on grid electricity with one backup generator capable of supporting essential functions for 24 hours. Chemical and spare part inventories typically maintain 2–3 weeks of operational buffer, enabling short duration continuity during access disruptions. However, no secondary water intake exists, and the system cannot operate if the primary intake is blocked by sediment or debris during extreme floods. Thus, adaptive capacity exists but is limited under prolonged hazard conditions."

Networks and linkages

- **Weak:** "Access roads to the facility are unpaved and vulnerable to wash outs, limiting reliable movement of staff and supplies after intense rainfall. Telecommunications rely on a single fiber line without redundancy; however, critical data systems have offline functionality. Power supply has only one feeder line. Overall, linkages have limited redundancy and would face prolonged disruption after flood or landslide events."

Outputs produced by the project

- **Moderate:** "Output continuity depends heavily on uninterrupted pumping and treatment processes. Existing SOPs allow temporary reduction in output while maintaining safety standards, and the facility can bypass nonessential processes under stress conditions. Output recovery after moderate flooding is typically achievable within 24–48 hours. However, severe disruptions to inputs or external systems would cause prolonged output losses."

What appraisers look for

- Clear link between each carried-forward hazard and the features that let the component continue or recover
- Materiality stated: can the component keep running, or how quickly can it recover at high severity?
- Evidence that ratings match current design and practices (existing, not planned)
- Consistency with 3.2a, 3.2b, 3.3a, and 3.3b.
- Justification that matches project documentation.
- Realistic evaluation that does not overstate capacity where vulnerabilities exist.

Scoring guidance

- **Pass:** Each component has a rating supported by specific features and buffers that explain performance at upper-bound severity and expected recovery time. Evidence based reasoning tied to actual design or operational features. Internal consistency with Q3.2a, Q3.2b, and Q3.3a.

- **Not Clear**: Justifications are generic, incomplete, or inconsistent. Features are listed but their sufficiency is vague. Ratings are plausible but unsupported. Materiality is not clear or only current conditions are considered. The distinction between current vs. planned features is blurred.

- **Fail**: Ratings are missing or unjustified. Responses lack explanation, rely on location rather than component capacity, or merely restate the label (for example, “capacity is high”) with no evidence. Capacity is overstated contrary to project documentation.

3.3 RISK MANAGEMENT FRAMEWORK

Section 3.3 captures how the project will reduce its climate and geophysical risks through (i) measures already integrated into the design, (ii) additional measures planned for future implementation, and (iii) approaches to manage any (residual) risks that remain after all feasible actions have been taken. This section links directly to the vulnerabilities identified in Q3.2 and forms the basis for climate-informed adjustments to the project's cost, design, and implementation arrangements.

CDRS Question 3.3a [Existing adaptation measures]: As currently designed, what measures have the project already incorporated to address the risks and vulnerabilities identified in Q3.2?

Purpose of the Question

This question identifies current ("already built-in") structural and non-structural measures that reduce exposure, sensitivity, or consequences of the hazards identified in Q3.2. These are features the project already includes at this stage of design, distinct from future measures (Q3.3b). This assessment helps appraisers determine whether the design has already internalized climate-risk good practice, and whether major design gaps remain.

Existing measures fall under three categories:

- ✓ **Structural measures**: Physical, engineered, or locational design features that lower hazard impacts.
- ✓ **Soft measures**: Policies, procedures, maintenance arrangements, plans, and institutional systems that improve resilience.
- ✓ **Other measures**: Any additional project-specific resilience feature not captured above.

How project planners should respond

Project planners are encouraged to follow this three-step process in order to produce credible and complete response to Q3.3a:

Step 1: Review each adaptation option listed in the NPP. The NPP lists three categories: (i) structural measures, (ii) soft measures, and (iii) other measures. For each row in the table, examine whether the project *already incorporates* that option. Select "Yes" only if it is clearly documented in the current design or existing institutional arrangements. Select "No" if the measure is not present *or* is merely planned or anticipated.

Common structural measures include:

- Elevating equipment above historical flood levels
- Wind-resistant roof trusses designed for cyclone-severity loads

- Fire-resistant materials in wildfire-prone areas
- Drainage sized for intense rainfall
- Slope stabilization, retaining walls, rockfall barriers
- Flexible joints, seismic reinforcement, etc.

Common soft measures include:

- Written maintenance plans for drainage, pumps, culverts
- Emergency operating procedures (EOPs)
- Real-time hazard monitoring, alarms, and early-warning integration
- Institutional capacity (trained operators, safety teams)
- Long-term planning frameworks that account for future climate changes
- Data systems that capture climate-related performance information

Other measures:

- Project-level contingency funds
- Community-based resilience mechanisms
- Integration with district-level disaster-response frameworks

Step 3: Provide a concise but specific justification. A strong justification must explain in 2–4 short but substantive sentences:

- The measure in place
- Which hazard/vulnerability it addresses,
- How it reduces risk or enhances resilience,
- Whether it introduces any maladaptation risks,
- How it interacts with the wider system (e.g., district drainage, utility networks, upstream hydrology, workforce access, institutional capacity).

The system-level assessment asks whether the measure works effectively with related infrastructure, service chains, supply systems, or institutional arrangements. For example, does the drainage connect to functional municipal drains, or will it fail without upstream improvements? Does an on-site backup generator depend on a fuel supply chain that may itself be disrupted? Does slope stabilization protect the site but increase landslide risk downstream?

For the maladaptation check, explicitly consider whether the measure might unintentionally increase vulnerability elsewhere, worsen downstream impacts, lock the project into rigid infrastructure that performs poorly under future climate scenarios, or reduce flexibility or adaptive room over the life of the asset. If no maladaptation risks exist, briefly state that and why.

Minimum evidence required

- A clear description of the existing measure
- A direct link to at least one hazard from Q3.2
- An explanation of how the measure reduces risk
- Evidence drawn from concept designs, feasibility studies, or standard operating procedures (where available)

If the measure is not yet confirmed or only "expected," it belongs in Q3.3b. Do not evaluate exposure or sensitivity again. Q3.3a explains only what the measure does, not why the hazard matters or whether the component is sensitive. Keep the text factual, concise, and limited to existing measures. Do not mention future enhancements or aspirational plans here.

What appraisers look for

PASS if:

- The response identifies specific, credible, already-integrated measures,
- Measures directly match vulnerabilities identified in Q3.2,
- No obvious gaps exist between hazards rated Moderate/High and claimed measures.

NOT CLEAR if:

- Measures are listed but links to hazard vulnerabilities are weak, incomplete, or generic,
- The response lists intentions or aspirations rather than proven existing measures,
- Some vulnerabilities appear unaddressed.

FAIL if:

- No measures are provided when vulnerabilities exist,
- The response is generic or contradictory,
- Measures listed do not actually reduce the identified hazards or appear fabricated.

Screenshot of sample justifications for Q3.3a (planners may adapt)

Measure	Yes / No	Capture your analysis here
(i) Structural measures: Physical or engineered changes to the design of the project or its location specifically to reduce hazard impacts	Yes	The main building and pump house are elevated 0.8 m above surrounding grade, reducing the impact of the High flood exposure identified in Q3.2a. These elevations align with district flood maps and maintain connectivity to municipal drainage. No maladaptation risk is identified because the raised platform does not divert floodwater to adjacent communities. Within the broader system, effectiveness depends on continued maintenance of upstream drainage channels.
(ii) Soft measures: Policy, operational, maintenance, or monitoring arrangements that support long-term risk management. *(Please indicate whether each measure has already been incorporated into the project's current design and summarize your analysis.)*		
1. Policy Development	No	
2. Emergency Planing	Yes	An emergency operating procedure (EOP) is already established for severe rainfall events, including controlled shutdown and safe evacuation protocols. This reduces sensitivity to pump failure risks identified in Q3.2b. The EOP aligns with district disaster-response frameworks and does not create maladaptation concerns because it coordinates with external early-warning systems rather than relying solely on on-site triggers.
3. Long-Term Strategic Planning	No	
4. Maintenance and Operations	Yes	A preventive maintenance schedule for desilting drains, reinforcing culverts, and inspecting retaining walls is already embedded in project design. This reduces vulnerability to landslide and heavy-rainfall hazards flagged in Q3.2. The approach supports system-level resilience by keeping onsite drainage functional within the district network. No maladaptation risks identified.
5. Capacity Building, Training, and Outreach	No	
6. Data Gathering, Monitoring, and Information Systems	No	
(iii) Other measures (Specify)	Yes	The project incorporates a 5% contingency fund earmarked for climate-related repair costs, supporting faster recovery from hazard impacts and improving adaptive capacity.

CDRS Question 3.3b [Planned Adaptation Measures]: What additional measures, if any, will the project take to adapt to the climate and geophysical hazards identified in Q3.2a that the project location may face in the future?

Purpose of the Question

This question identifies future climate- and disaster-risk reduction measures that the project team commits to integrate during detailed design, construction, or operation. It builds directly on the vulnerabilities identified in Q3.2 and helps appraisers determine:

- Whether remaining risks (after existing measures in Q3.3a) are addressed,
- Whether proposed actions are appropriate, feasible, and proportional to the hazard profile,
- Whether planned measures avoid maladaptation, and
- Whether the project's resilience is strengthened within the broader system (e.g., hydrology, utility networks, institutional frameworks).

A strong response demonstrates that the team understands the project's vulnerabilities and is taking targeted, evidence-based action to manage them.

How project planners should respond

Follow the same logic as 3.3a but with these key differences:

- **Focus only on remaining gaps**. Identify the risks from Q3.2 that still need attention because existing measures (3.3a) do not sufficiently address them.

- **Propose clear, actionable planned measures**. Describe structural, soft, or operational measures that will be added to strengthen resilience and explain the benefit of each measure. Briefly state which hazard the measure addresses and how it reduces exposure, sensitivity, or downtime.

Screenshot of sample justifications for Q3.3b (planners may adapt)

3.3b [PLANNED ADAPTATION MEASURES] What additional measures, if any, will the project take to adapt to the climate and geophysical hazards identified in Q3.2a that the project location may face in the future? *(Please describe any planned structural improvements, soft measures, or operational adjustments that will be implemented during construction or operation to strengthen the project's resilience to future climate - and geophysical-related impacts.)*
To address High flood exposure and low adaptive capacity of electrical equipment, the project will raise the equipment platform by an additional 0.6 m during detailed design. This reduces inundation risk under projected future flood levels. The measure aligns with district drainage and does not divert water to adjacent properties (avoids maladaptation).

What appraisers look for

- Measures are appropriate and realistic.
- No duplication of existing measures from 3.3a.
- Proposed actions correspond to vulnerabilities identified in 3.2.

CDRS Question 3.3c: How will the project manage any residual risks that remain after applying existing and planned adaptation measures?

Purpose of the Question

This question assesses how the project intends to manage the climate and geophysical risks that remain even after all existing measures (Q3.3a) and planned measures (Q3.3b) have been accounted for. Residual risks typically arise from:

- Extreme events that exceed design thresholds,
- Dependencies on external systems (e.g., access roads, power supply, upstream drainage),
- Geophysical hazards that cannot be engineered away entirely,
- Uncertainty in long-term climate projections.

The project team must indicate which residual-risk strategy (avoidance, minimization, or compensation/risk transfer) will be employed and provide a concise justification for each selection.

How project planners should respond

The response for Q3.3c follows the same structured logic as Q3.3a and Q3.3b but focuses only on risks that remain after all adaptation measures have been integrated. To complete the table, the project team should:

Step 1: Identify remaining residual risks. The team should review Q3.2a–c and the measures selected in Q3.3a–b to determine which hazards still present meaningful risk. Residual risks should be clearly connected to the earlier vulnerability assessments and the limits of the existing and planned adaptation measures.

Step 2: Select the most appropriate risk-management strategy for each option. Each strategy in the NPP table reflects a different risk posture:

- *Avoidance* is appropriate when risks can be eliminated by relocating the project, redesigning core components, or changing technologies.

- *Minimization* applies when risks cannot be entirely removed but can be reduced to tolerable levels through additional procedures, standards, or contingency arrangements.

- *Compensation / Risk Transfer* is used as a last resort to off-set impacts. It is suitable when risks cannot be reduced further and must be absorbed or transferred (e.g., insurance, budget contingencies, emergency finance mechanisms).

- *Not Applicable* is selected only when no material residual risks remain after Q3.3a and Q3.3b.

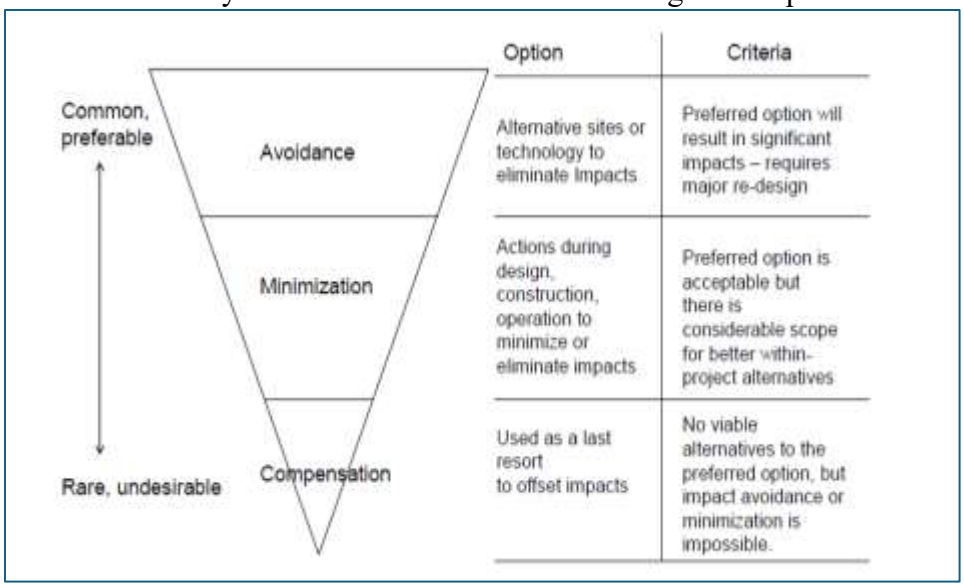
Step 3: Provide a brief justification for each "Yes" selection. Justifications should address:

- Which residual hazard(s) remain (e.g., rare but severe flooding, extreme rainfall events, low-probability earthquakes, systemic cascade failures),

- Why the chosen strategy is appropriate,

Justifications should be short (2–4 sentences) but analytically robust.

Taxonomy of climate and disaster risk management options



What appraisers look for

A strong submission demonstrates that the project team:

- Clearly understands the distinction between planned adaptation (Q3.3b) and residual-risk management,
- Identifies genuine residual risks that cannot be feasibly eliminated,
- Selects an appropriate risk strategy consistent with the project's context,
- Provides a hazard-specific, system-aware justification,
- Avoids unrealistic or contradictory claims (e.g., claiming "Not Applicable" when significant residual risks remain).

Pass: Strategies are well-reasoned, hazard-specific, and consistent with earlier assessments.

Not Clear: Reasoning is vague, generic, or inconsistent.

Fail: Residual risks are misidentified or not addressed; justifications are missing.

Box 5: Examples of justification statements

Avoidance: "*The project team identifies a residual landslide risk along the steep original alignment that cannot be eliminated through feasible stabilization. To fully avoid this risk, the alignment will be shifted upslope to an area with low susceptibility ratings. This adjustment prevents risk accumulation downstream and aligns with district land-use guidance.*"

Minimization: "*Even after drainage upgrades, the project remains exposed to rare extreme rainfall events capable of causing temporary access disruption. The project team therefore commits to implementing a contingency operations plan and maintaining emergency access protocols to minimize service interruption. These measures function within the broader district emergency-response system and do not introduce maladaptation*."

Compensation / Risk Transfer: "*Residual earthquake risk persists despite structural reinforcement. The project will secure insurance coverage for structural and equipment damage and establish a contingency reserve to support rapid recovery. This approach ensures financial resilience without encouraging costly over-design*."

Not Applicable: "*Following the elevation of critical assets, reinforcement of slopes, and integration with early-warning systems, the project has no remaining material residual risks requiring additional avoidance, minimization, or risk-transfer strategies. All identified hazards have been sufficiently addressed*."

3.4 PROJECT IMPACT ON THE CLIMATE AND ENVIRONMENT

This subsection of the NPP examines how the project itself affects the climate, the environment, and vulnerable groups, complementing the earlier analysis of how climate hazards affect the project (Q3.2–Q3.3). The questions that follow (Q3.4–Q3.7) require project teams to assess whether project activities may generate greenhouse gas emissions, alter natural systems, increase or reduce climate-related environmental impacts, or influence the vulnerability of specific population groups already at risk from climate change.

Together, these questions ensure that public investments do not unintentionally increase Malawi's emissions profile, degrade ecosystems, or exacerbate risks faced by vulnerable communities. They also give consideration to, and documents any positive mitigation, adaptation, or resilience co-benefits.

The information collected here also supports Malawi's national reporting obligations under the NDC, the National Climate Change Management Policy, and budgeting processes. Project teams should therefore provide specific, evidence-based descriptions of potential impacts, including areas of uncertainty requiring further investigation.

CDRS Question 3.4: What climate-related impacts can be expected if the project is implemented as planned?

Purpose of the Question

This question determines whether the project, as designed, is likely to generate climate-related impacts (positive or negative) during construction or operation. The analysis supports both climate-risk management and Malawi's national climate-reporting commitments. Unlike Questions 3.2 and 3.3, which focus on the project's vulnerability to climate hazards, Question 3.4 assesses the project's own contribution to climate change impacts (e.g., greenhouse gas emissions, changes to natural systems, or resilience-enhancing effects).

The project team must review each listed impact category in the NPP and select the appropriate response ("Yes," "No," or "Unknown"), followed by a short justification.

How project planners should respond

The response in Q3.4 is simpler and more declarative than the vulnerability and risk-management questions that precede it. The team should:

Step 1: Evaluate each impact category independently. For each statement (e.g., net change in GHG emissions, land-use change, hydrological impacts), the project team should:

- Review the project's activities and infrastructure footprint,
- Consider construction emissions, operational emissions, and changes to natural systems,
- Identify whether the project is likely to generate the impact,
- Select Yes, No, or Unknown accordingly.

Selections must reflect the current project design—not aspirational mitigation measures that may be added later.

Step 2. Provide a concise, factual justification. Each justification should briefly state:

- Which project activity creates (or avoids) the impact,
- Whether the effect is material (e.g., significant vs. negligible),
- Any known uncertainties (for "Unknown"),
- How the conclusion aligns with Sections 1 and 2 of the NPP (e.g., project scope, size, technology type).

Justifications should be project-specific and avoid generic statements such as "construction generates emissions".

Step 3. Distinguish between construction-phase and operational impacts. Where relevant, the justification should clarify whether the impact arises:

- During construction (e.g., earthworks, transport of materials, cement use),
- During operation (e.g., energy use, pump electricity demand, transport fleet),
- Both phases, where applicable.

If effects differ substantially between phases, this should be briefly noted.

Step 4. Consider system-level effects where relevant. Some categories, such as hydrological impacts, land-use change, or changes in local microclimates, require understanding of the broader system (river basin behaviour, ecosystems, catchment dynamics). The justification should acknowledge system interactions where they materially influence the impact assessment.

Step 5. Use "Unknown" only when evidence is genuinely unavailable. This option is appropriate if:

- The project team lacks data to make a reasonable assessment,
- Key design elements are not yet defined,
- Impact magnitude depends on future modelling or specialist studies.

Where "Unknown" is selected, the justification must indicate:

- Why the information is unavailable, and
- What additional work (if any) will be undertaken to resolve the uncertainty.

What appraisers look for

- No contradictory responses.
- Justification demonstrates understanding of the project's emissions profile and resilience implications.
- Internal consistency with Sections 3.1 and 3.5.

Screenshot of sample justifications for Q3.4 (planners may adapt)

Measure	Response	Capture your justification here
The project qill result in:		
a) A net change in greenhouse gas (GHG) emissions over its lifecycle (construction and operation).	Moderate Increase: 5,000–20,000 tCO_2e/yr	Construction will involve cement-intensive works and transport of aggregates, generating short-term GHG emissions. During operation, electricity demand from pump systems will contribute to indirect emissions. The magnitude is expected to be moderate relative to national baselines.
b) Significant GHG emissions from construction, energy use, fuel combustion, transport, or industrial processes.	No	The project is located entirely within an existing serviced industrial plot and requires no vegetation clearance or disturbance of natural carbon sinks.
c) Emissions of carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), or fluorinated gases.	Yes	The project modifies stormwater runoff patterns by paving 0.5 ha of previously permeable surface, which could increase downstream peak flows. Appropriate mitigation will be evaluated in detailed design.
d) The removal or degradation of natural carbon sinks or significant land-use change (e.g. forests, wetlands, soils, mangroves).	No	No clearing of natural areas
e) Changes in hydrological patterns or natural systems that could increase flood risk, drought risk, or water stress.	Unknown	The project may require excavation in areas with undocumented subsurface materials. A geotechnical and environmental assessment is planned to determine whether construction could affect natural drainage or soil carbon.
g) Changes in local microclimate or ecosystems that could affect climate resilience (e.g. heat release, biodiversity loss).	No	No habitat change, small footprint
h) The lock-in of high-carbon infrastructure or the acceleration of a transition to low-carbon technologies or systems.	No significant effect	The project does not rely on long-lived infrastructure or technologies that would commit it to carbon-intensive operations over time.
Other (specify)	No additional impacts	

CDRS Question 3.5: What strategies are in place to reduce the climate-related impacts of the project that were identified in Q3.4?

Purpose of the Question

This question screens whether the project incorporates any mitigation measures—that is, actions that reduce or avoid greenhouse gas (GHG) emissions during construction or operation. Unlike Section 3.3, which focuses on adaptation, Section 3.5 assesses low-carbon opportunities that the project may already include, either directly or indirectly. The aim is to identify positive climate-mitigation contributions without requiring a full GHG inventory or detailed engineering analysis.

How project planners should respond

To complete Q3.5, the project team must follow the two-step logic required by the NPP:

Step 1: Determine whether any mitigation strategies are in place. The team should review the current project design and confirm whether it already incorporates any measures that reduce greenhouse gas emissions during construction or operation. If no mitigation strategies are present, the team selects "No" and provides a brief justification (1–2 sentences). If one or more mitigation strategies are present, the team selects "Yes."

Step 2: If "Yes," select all mitigation strategies that apply. If the team selected "Yes" in the previous step, they must review the NPP's eight predefined strategy types and tick all that accurately describe the project's GHG-reducing features. For each selected strategy, the team should provide a short, project-specific justification explaining how the measure reduces emissions or avoids generating new ones. Justifications should be concise and should not duplicate adaptation measures already described in Section 3.3.

Screenshot of sample justifications for Q3.5 (planners may adapt)

Measure	Response	Capture your justification here
Are any mitigation strategies in place? (if yes, select from below all that apply)	Yes	
a) Energy-efficiency measures integrated (e.g., efficient equipment, reduced energy demand)	Yes	The project includes high-efficiency pumps that reduce electricity demand relative to standard equipment, resulting in lower operational GHG emissions.
b) Low-carbon or renewable energy incorporated (e.g., solar for facilities, clean energy for operations)	No	The project does not incorporate renewable-energy systems, as the scale and functional requirements make such integration technically or financially unfeasible at this stage.
c) Low-emission construction or materials used (e.g., reduced-carbon cement, minimal earthworks)	Yes	The design specifies reduced-carbon cement in structural works, lowering embodied emissions associated with construction.
d) Design changes reduce emissions sources (e.g., smaller footprint, optimized routing, process redesign)	No	The project's design parameters do not allow for further footprint reduction or routing optimisation beyond what is required for core functional performance.
e) Land-use or ecosystem protection included (avoids emissions from land clearance or supports sequestration)	No	The project is located within an existing serviced site and does not involve land clearance or disturbance of natural carbon sinks.
f) Operational practices to reduce emissions (e.g., efficient fleet management, maintenance protocols)	Yes	The project incorporates operational protocols that reduce emissions, including scheduled equipment maintenance to maintain energy efficiency and minimise fuel or electricity waste over the asset's lifecycle.
g) Waste or resource-efficiency measures added (materials reduction, improved waste handling)	Yes	The project incorporates a materials-reduction plan during construction, including reuse of excavated material where feasible, helping reduce waste-related emissions.
h) Other (specify)	Yes	The project includes digital management tools to optimise system performance, reducing unnecessary energy consumption and indirectly lowering operational emissions.

What appraisers look for

- Selected options must be supported by actual project features.
- Explanations should be consistent with mitigation claims in Section 3.4.
- No copying from Section 3.1b.

CDRS Question 3.6: What other environmental impacts have been identified in the project proposal and what further investigation on mitigations is planned?

Purpose of the Question

This question identifies non-climate environmental impacts associated with project construction and long-term operation, and documents any additional investigations or assessments planned to better understand these impacts and guide mitigation. These may include impacts on soils, water quality, drainage patterns, habitats, noise, dust, waste, construction footprint, erosion, or downstream communities. This is distinct from Q3.4 (GHG emissions) and Q3.5 (GHG mitigation). Here the focus is on local environmental harm and how the project will study and manage it.

How project planners should respond

To complete Q3.6, the project team must follow this four-step process:

Step 1: Identify likely environmental impacts from the project throughout its lifecycle. The team must describe impacts from c**onstruction** (e.g., excavation, spoil disposal, noise, dust, short-term water pollution, soil disturbance, changes in runoff patterns) and o**peration** (e.g.,

wastewater generation, erosion, changes to hydrology, solid waste, noise, heat release, increased runoff). Impacts must be specific, credible, and linked to the project's physical activities and scale.

Step 2: Explain why these impacts are likely. Provide a brief, factual explanation of the mechanism.

Step 3: Describe any planned environmental studies or assessments. The NPP requires evidence of **forward-looking mitigation planning**, which may include Environmental and Social Impact Assessment (ESIA)**.** Indicate the **focus** of any follow up study, not just its title.

Step 4: Keep the response aligned with the project scale. Do not overstate or understate impacts. The analysis should reflect the nature of works, footprint size, proximity to sensitive areas, operational processes, and expected intensity and duration of impacts.

Mistakes to avoid

- Restating GHG impacts already covered in Q3.4
- Providing generic statements ("There may be some environmental impacts…")
- Omitting operational impacts (long-term effects matter)
- Listing studies without specifying what they will investigate
- Failing to mention impacts that are normal and expected for the project type (e.g., dust, sediment, excavation waste)

Screenshot of sample write-up for Q3.6

3.6 What other environmental impacts have been identified in the project proposal and what further investigation on mitigations is planned? *(Most construction projects cause some environmental harm, describe these impacts. Also describe any environmental harm that might result from the long-term operation of the project. Describe any planned studies or impact assessment work aimed at minimizing harm)*
Construction is expected to generate dust, noise, excavation waste, and increased sediment runoff, and may temporarily alter drainage patterns until final grading is complete. During operation, the project could slightly increase impermeable surfaces and produce small amounts of sludge and wastewater requiring controlled disposal. To address these issues, a geotechnical and environmental assessment will be carried out, alongside stormwater-management analysis and a construction environmental management plan. These studies will guide specific mitigation actions to minimise environmental harm.

What appraisers look for

- Information is realistic and proportional to project size.
- Impacts are not overlooked, downplayed, or generalized.

CDRS Question 3.7: How will the project affect vulnerable communities, particularly those that are already at risk from climate change?

Purpose of the Question

This question assesses whether the project will increase, reduce, or have no material effect on climate-related risks faced by vulnerable communities. Vulnerable communities include groups whose livelihoods, health, housing, or safety are already compromised by climate hazards such as flooding, drought, extreme heat, landslides, or storms. These may include:

- Low-income households in hazard-prone areas,
- Communities dependent on climate-sensitive livelihoods (e.g., rain-fed farmers),

- Populations living in informal settlements,
- Elderly, persons with disabilities, and female-headed households,
- Communities with limited access to services, emergency response systems, or infrastructure.

The purpose is to ensure that public investments do not unintentionally worsen climate vulnerability and, where possible, contribute to risk reduction and enhanced resilience.

How project planners should respond

Project teams must select one or more of the four options provided in the NPP Template and include a short, evidence-based justification for each selected statement. Responses must draw on the project's location, design, beneficiary profile, and the hazard trends identified in Sections 3.2. Follow these four steps:

Step 1: Identify the affected community groups. Determine whether the project affects:
- Communities in or near the project site,
- Communities depending on services provided by the project,
- Communities downstream, upland, or otherwise connected through hydrology, transport, or service networks.

Examples:
- A water project affecting downstream households through changes in drainage patterns,
- A road upgrading project affecting informal settlements located along the corridor,
- A health facility project serving populations vulnerable to heat and vector-borne diseases.

Step 2: Assess the mechanism of impact. Assess whether the project will increase, reduce, or have no meaningful impact on vulnerability, then select the statements that apply. Present uncertainty (only when evidence is not yet available). Multiple selections may apply only if clearly justified.

Step 3: Provide a concise justification. Each justification must be 2–4 sentences and include:
- The specific community affected,
- The relevant hazard(s),
- The mechanism of impact (positive, negative, or neutral),
- The evidence used (local knowledge, assessments, hazard maps, exposure analysis).

Avoid generic statements such as "the project benefits everyone equally" or "no negative effects expected." All claims must be linked to project activities.

Screenshot of sample write-up for Q3.7. These examples illustrate the expected level of detail. Replace with project-specific information.

3.7 How will the project affect vulnerable communities, particularly those that are already at risk from climate change?		
Likely impact	**Response**	**Capture your justification here**
a) The project is likely to **increase** vulnerable communities' exposure or sensitivity to climate hazards, or to reduce their adaptive capacity.	Yes	The project will increase surface runoff due to expanded paved areas, which may raise flooding risk for low-income households located immediately downstream. Hazard maps indicate that the settlement lies within a moderate flood-exposure zone, and current drainage systems are inadequate. Without complementary measures, the project may unintentionally heighten climate-related risks for these communities.
b) The project is likely to **reduce** vulnerable communities' exposure to climate hazards or strengthen their adaptive capacity.	Yes	The project includes the rehabilitation of drainage channels and reinforcement of culverts along a road corridor serving flood-prone communities. These measures directly reduce peak-flow flooding and ensure all-weather access during heavy rainfall events. As a result, households already vulnerable to climate hazards will benefit from reduced exposure and improved access to essential services
c) The project is **unlikely to affect** vulnerable communities' exposure, sensitivity, or adaptive capacity.	Yes	The project site is situated in an industrial zone with no residential communities nearby. It does not alter drainage, service access, or hazard pathways that affect vulnerable populations. Therefore, the project is unlikely to materially change climate risks for surrounding communities.
d) The project's impact on vulnerable communities is **uncertain** due to insufficient information at this stage.	Yes	The project may influence hydrological patterns, but baseline data on downstream settlements and drainage pathways is currently unavailable. A hydrological and social-risk assessment will be conducted during detailed design to determine whether vulnerable communities could be affected. Until these assessments are completed, the project's impacts cannot be conclusively determined.

What appraisers look for

Appraisers assess:

- Whether the selected statements reflect evidence from Sections 3.2 and 3.4
- Whether justifications are specific, clear, and non-generic
- Whether hazard pathways and community vulnerabilities are logically explained
- Whether "uncertain" is used appropriately (not as a default)
- Whether selections contradict each other

Scoring guidance

Pass: The selected option(s) are credible and supported by clear, project-specific justification.

Not Clear: Statements are plausible but lack sufficient detail, precision, or evidence.

Fail: No explanation provided, statements conflict with earlier analysis, or claims are generic and unsupported.

CDRS Question 3.8: What climate change indicators will be monitored to track the project's performance and identify potential impacts?

Purpose of the Question

This question ensures that each project includes specific, measurable indicators to track climate-related performance throughout its lifecycle. These indicators:

- Demonstrate whether adaptation or mitigation measures are functioning as intended,
- Help MDAs and the PSIP Unit monitor climate-related risks and benefits,
- Support national reporting to the NCCMP, NDC, and Malawi's budgeting systems,
- Allow early detection of negative impacts or emerging risks.

Indicators must follow the SMART principle: *Specific, Measurable, Achievable, Relevant, and Time-bound*. They should be directly linked to the project's design, hazards identified in Q3.2, adaptation measures in Q3.3, and climate-related impacts in Q3.4–3.5.

How project planners should respond

Project planners must propose **2–5 climate-related indicators**, depending on the nature and scale of the project. Indicators should be:

- **Actionable**: clearly tied to project-level interventions,
- **Relevant**: linked to adaptation or mitigation claims made earlier,
- **Trackable**: measurable with existing systems or simple additions,
- **Feasible**: within the implementing agency's monitoring capacity.

Teams are advised to follow these steps:

Step 1: Identify the climate-related elements of the project. Draw directly from:

- Exposure, sensitivity, and adaptive-capacity findings (Q3.2),
- Existing and planned adaptation measures (Q3.3a, Q3.3b),
- Emission-related findings (Q3.4–3.5),
- Vulnerable community impacts (Q3.7).

Ask: *Which climate risks or benefits require ongoing monitoring to ensure the project remains resilient and low-carbon?*

Step 2: Select indicators that are meaningful and measurable. Indicators may track:

- Hazard conditions (e.g., rainfall intensity, groundwater levels),
- Performance of adaptation measures (e.g., drainage capacity, emergency-response activation time),
- Impacts on operations or communities (e.g., service disruptions during heavy rainfall),
- GHG-related outcomes (e.g., electricity consumption, fuel use),
- Institutional performance (e.g., frequency of maintenance activities relevant to climate resilience).

Indicators must reflect variables that the project can realistically monitor using existing tools, logs, or reports, and should not require advanced modelling.

Step 3: Ensure indicators are SMART and directly linked to project logic. Each indicator should clearly state:

- What is monitored (e.g., drainage flow rates),
- How it will be measured (e.g., using monthly O&M logs),

- Target or threshold where applicable (e.g., no more than 2 flood-related interruptions per year),
- Frequency of measurement (e.g., quarterly, annually).

Avoid overly broad indicators like "climate resilience improved" or "flooding reduced." Indicators must be quantitative or binary, not aspirational.

Box 6: Sample Indicators (for illustration)
(Replace these with project-specific indicators tailored to the design, sector, and hazards.)

Adaptation-focused indicators

- Number of flood-related service interruptions per year, measured through operations logs.
- Percentage of culverts/drainage structures cleared according to schedule, based on maintenance records.
- Maximum flood depth recorded at the project site per rainy season, using simple staff-gauges or digital sensors.
- Average time required to restore service after extreme-weather disruptions, using emergency-response logs.
- Temperature-related equipment downtime, tracked using equipment failure logs.

Mitigation-focused indicators

- Annual electricity consumption (kWh) relative to baseline design assumptions.
- Fuel consumption of project vehicles or generators, measured by procurement or O&M records.
- Tonnes of reduced-carbon materials purchased, based on procurement invoices.

Indicators for vulnerable-community impacts

- Number of community complaints related to drainage or runoff changes, collected via grievance-redress systems.
- Households benefiting from reduced climate-risk exposure, using beneficiary monitoring.

These examples illustrate expected detail. Planners should adapt them to the project's context.

Mistakes to avoid

- Providing vague indicators (e.g., "improve resilience").
- Selecting indicators unrelated to the project's hazards, adaptation measures, or emissions profile.
- Proposing monitoring activities beyond the agency's realistic capacity.
- Omitting indicators for claimed adaptation or mitigation benefits.

What appraisers look for

Appraisers assess whether indicators:

- Are specific, measurable, and clearly linked to earlier findings (Q3.2–3.7),
- Reflect climate-relevant variables that the project can actually monitor,
- Are realistic given project capacity and sector standards,
- Are not generic (e.g., "climate change indicator TBD"),
- Avoid duplication unless justified.

Internal inconsistencies are red flags, e.g., claiming adaptation benefits in Q3.3 but offering no adaptation indicators.

Scoring guidance

- **Pass**: Indicators are SMART, climate-relevant, and connected to project-level risks or adaptation/mitigation measures.

- **Not Clear:** Indicators are listed but are generic, immeasurable, or poorly aligned with earlier sections.

- **Fail**: No indicators provided; indicators irrelevant to climate risks; or indicators do not relate to project design or monitoring systems.